Problem 1

B C E F

Problem 2

(a) The inverter is faulty and acting as a short (the gate acts as a standard AND gate).

(b) faulty cause gate outputs $A\overline{B}$ instead of NAND

(c) the gate ignores B

(d) The gate outputs $\overline{A}B$ instead of XOR

Problem 3

(a) $\overline{AB}$

(b) $A + \overline{B}$

(c) $\overline{ABC}$

(d) $\overline{A} + \overline{B} + \overline{C}$

(e) $A(B + C)$

(f) $(\overline{A} + \overline{B})(\overline{C} + \overline{D})$

(g) $(A + \overline{B})(\overline{C} + \overline{D})$

(h) $(A + \overline{B})(\overline{C} + D)$

Problem 4

(a) $X = AB + C$

(b) $X = A + \overline{B}$

(c) $X = ABCD$

(d) $X = (A + B + C)D$

Problem 5

(a) Connect B and C to an OR gate, then put that output and A into an AND gate.

(b) Make an AND gate for A and B and a 3-input AND gate for A, B, and C then connect both of their outputs into an OR gate.

(c) Connect A, B, and C to a 3-input NAND gate then put that output and D into an OR gate.

(d) Connect B, C, and D to a 3-input OR gate then put that output and B into an AND gate then connect that output and A into an OR gate.

Problem 6

(a) A, B: 00 → 0, 01 → 1, 10 → 1, 11 → 1

(b) A, B: 00 → 0, 01 → 0, 10 → 0, 11 → 1

(c) X=1 if A=1. If A=0, X=1 only when B=1 AND (C=1 OR D=1). All other combinations output 0.

(d) X=0 only when A, B, C are all 0

(e) X=1 for inputs 011, 110, 111

Problem 7

Note: None of the circuits are equivalent.

(a) $X = AB + CD$

(b) $X = A(B + C + D)$

(c) $X = AB$

(d) $X = A(B + CD)$

Problem 8

(a) X=1 for inputs 011, 110, 111
(b) X=1 for inputs 010, 011, 100, 101, 111

Problem 9

(a)

SOP: $\overline{A}\overline{B}C + A\overline{B}\overline{C} + A\overline{B}C + ABC$

POS:
$(A + B + C)(A + \overline{B} + C)$
$(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C)$

(b)

SOP: $A\overline{B}C + AB\overline{C} + ABC$

POS:
$(A + B + C)(A + B + \overline{C})(A + \overline{B} + C)$
$(A + \overline{B} + \overline{C})(\overline{A} + B + C)$

(c)

SOP: $\overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D} + A\overline{B}\overline{C}D + AB\overline{C}\overline{D}$

POS:
$(A + B + \overline{C} + D)(A + \overline{B} + C + D)(A + \overline{B} + \overline{C} + \overline{D})$
$(\overline{A} + B + C + D)(\overline{A} + B + \overline{C} + D)(\overline{A} + B + \overline{C} + \overline{D})$
$(\overline{A} + \overline{B} + C + \overline{D})(\overline{A} + \overline{B} + \overline{C} + D)(\overline{A} + \overline{B} + \overline{C} + \overline{D})$

(d)

SOP: $\overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}BCD + A\overline{B}CD + ABCD$

POS:
$(A + B + C + D)(A + B + C + \overline{D})(A + B + \overline{C} + \overline{D})$
$(A + \overline{B} + \overline{C} + D)(\overline{A} + B + C + D)(\overline{A} + B + C + \overline{D})$
$(\overline{A} + B + \overline{C} + D)(\overline{A} + \overline{B} + C + D)(\overline{A} + \overline{B} + C + \overline{D})$
$(\overline{A} + \overline{B} + \overline{C} + D)$

Problem 10

(a) $\overline{B}C + \overline{A}B\overline{C}$

(b) AC

(c) B

(d) $A\overline{C} + B\overline{C} + \overline{ABC}$